

Matthew Stuart Jones

847-431-7231 | matthew.stuart.jones@icloud.com | Portfolio Website | LinkedIn

EDUCATION

University of Kentucky

Bachelor of Science, Computer Science

Projected Graduation, May 2027

GPA: 4.0/4.0

- Minors: Mathematics and Vocal Performance
- Artificial Intelligence Certificate
- Study Abroad Experience in Bilbao, Spain (Universidad de Deusto)

RESEARCH EXPERIENCE

NSF Research Experience for Undergraduates (REU), AI for Social Good

Summer 2026

Worcester Polytechnic Institute (WPI) — Worcester, Massachusetts

- Developed ARISES, a novel feature selection method, outperforming all state-of-the-art baselines on 6 of 7 multi-label metrics
- Developed MLX, a counterfactual explainability method using structured gradient descent with label-co-occurrence-weighted gradients, outperforming SHAP and LIME on multi-label, patient-wise explanations for black-box classifiers
- Applied prompt engineering and AI tooling across the pipeline, matching tool to task for efficiency without losing accuracy

Independent Study: Responsible AI Citation & Policy Influence Mapping

Spring 2026

Advisor: Dr. Brent Harrison, University of Kentucky — Human Centered AI, AI Ethics

- Analyzed citation flows between RAI literature (AIES, NeurIPS) and 18 national AI governance frameworks; identified three-way misalignment between paper production (fairness), citation impact (explainability), and policy language (accountability)
- Built end-to-end data pipeline using Semantic Scholar API, intfloat/e5-large-v2 embeddings, and cosine similarity scoring to quantify research-to-policy alignment across a 20,000+ paper corpus
- Presented findings at the 20th Annual University of Kentucky Showcase of Undergraduate Scholars

AI Researcher for ACCESS

October 2024 – Present (incl. Summer 2025)

UK Center for Computational Sciences (CCS) — Lexington, Kentucky

- Built MCP-based semantic router to categorize incoming queries and route them to the appropriate AI agent across the ACCESS cyberinfrastructure ecosystem
- Developed Retrieval-Augmented Generation (RAG) system to assist users in drafting allocation abstracts and proposals based on a corpus of past submissions
- Designed multi-LLM tournament tagger to categorize research proposals by field of science using open-source models deployed on a Grace Hopper cluster via Ollama
- Improved existing AI models through document verification and retraining workflows

INVITED TALKS

Guest Lecturer, HON 251: Technology and the Human Experience (Honors)

Fall 2025

“A Brief Overview of AI Policy” — University of Kentucky

CONFERENCES

- “UK NSF Regional AI Workshop” — Lexington, Kentucky October 7, 2025
- “UK AI/ML Research Symposium” — Lexington, Kentucky March 28, 2024; November 12, 2024
- “UK Commonwealth Computational Sciences Summit on AI” — Lexington, Kentucky October 17, 2023

SKILLS

- **AI/ML Engineering:** Built RAG systems, MCP semantic routers, and LLM-based tagging pipelines using LlamaIndex, ChromaDB, sentence-transformers, and Ollama; experience with OpenAI API and local open-source models.
- **Mathematical Foundations:** Strong background in algorithms, linear algebra, probability, and numerical methods; coursework in sequential decision making, deep learning, and models of computation.
- **Programming & Systems:** Python (NumPy, pandas, scikit-learn) for ML/AI; C and C++; SQL; full-stack web development (JavaScript, HTML, CSS); Linux; R.
- **Research & Data Pipelines:** Designed end-to-end NLP pipelines for literature analysis at scale — API-based corpus

ingestion, embedding generation (e5-large-v2), and cosine similarity scoring across large document corpora; familiar with academic literature APIs (Semantic Scholar, OpenAlex).

- **Policy Communication:** Translates complex AI and technical concepts for policy, academic, and general audiences; experience analyzing national AI governance frameworks across multiple jurisdictions; public speaker and technical presenter; fluent in Spanish.

FEATURED COURSEWORK

CS 667	Sequential Decision Making	CS 463G	Intro to AI
CS 575	Models of Computation	CS 315	Algorithms
CS 485G	Data Ethics	PPL 306	Ethics and Civic Leadership
MA 322	Matrix Algebra and Its Applications	MA 321	Intro to Numerical Methods
MA 320	Introductory Probability		

HONORS AND AWARDS

- Dean's List — 6 Semesters
- Tau Beta Pi Scholar
- Pigman College of Engineering Marky Campbell Scholarship
- UK Bluegrass Spirit Scholarship
- UK Education Abroad Scholarship

PROFESSIONAL ORGANIZATIONS

Kentucky Alpha Chapter of Tau Beta Pi

Tau Beta Pi Scholar & Chapter Treasurer

- Managed chapter finances for a 120-member engineering honor society, including dues, budgeting, and scholarship requests
- Coordinated food and logistics for chapter meetings and events
- Volunteer judge at Future Cities regional engineering competition

ORGANIZATIONS AND INVOLVEMENT

acoUstiKats A Cappella

Business Manager & Social Media Manager

- Led the group's public presence and built an automated booking pipeline syncing forms with Google Calendar
- Spearheaded the group's first ICCA campaign where I initiated the decision to compete and led competition research
- Created content across platforms, designed and produced group merchandise, and publicized campaign events

UK Men's Chorus

Baritone Section Leader

- Serve as section leader on the group's New York tour, coordinating logistics and welfare for the baritone section
- Part of a leadership team focusing on culture and performance where singers are focused while taking care of each other

PUBLICATIONS

"The Translation Between Responsible AI Research and AI Legislation"

May 2026

20th Annual University of Kentucky Showcase of Undergraduate Scholars

"Predicting Adverse Reactions and Identifying Clinically Meaningful Features in Oncology Patients Using the US FAERS"

In Progress

Poster — NSF REU in AI for Social Good, Worcester Polytechnic Institute

Summer 2026

Matthew Jones*, Amruta Soma*, Gandhali Sheth, Dr. Chun-Kit Ngan, Dr. Bo Wang, Dr. Yin Ting Cheung

*Co-First Authors

"Predicting Adverse Reactions and Identifying Clinically Meaningful Features in Oncology Patients Using the US FAERS"

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Paper — NSF REU in AI for Social Good, Worcester Polytechnic Institute

Summer 2026

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REFERENCES

Dr. Brent Harrison

Director of Undergraduate Studies, CS
Pigman College of Engineering, UK
brent.harrison@uky.edu
(859) 257-8040

Dr. Mike Johnson

Tau Beta Pi Faculty Advisor
University of Kentucky
mike.johnson@uky.edu
(859) 257-0717

Dr. Ruiquan Huang

Assistant Professor, Department of CS
University of Kentucky
ruiquanhuang@uky.edu
(859) 257-3391

Dr. Jim Griffieon

Director, UK Center for Computational Sciences
University of Kentucky
griff@netlab.uky.edu
(859) 257-6746

Dr. Cory Curl

Director of Undergraduate Studies
Martin School of Public Policy, UK
cory.curl@uky.edu
(859) 562-3182